AI Course: Mini Project

Assignment 2: Real-Life Use Cases of Reinforcement Learning (RL)

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Project Objective

This mini-project focuses strictly on formulating and solving a practical problem using **Reinforcement Learning (RL)**. With idea submissions now closed, all approved teams may proceed with the implementation of their selected topics.

Phase 3: Project Submission Schedule

Date & Time	Phase	Action Required
Oct. 26, 2026, 10:00 AM (Sharp)	Final Phase	Final deadline for project submission and report.

- **Strict Deadline:** The submission portal will close promptly at 10:00 AM on October 26, 2026. No late submissions will be accepted.

Project Deliverables

This assignment is **strictly restricted to Reinforcement Learning**. Heuristic search, supervised methods, or non-RL approaches will be disqualified. Each team must submit:

1. **GitHub Repository Link:**
   - Must contain the complete, runnable source code, environment setup, and trained RL agent.
   - Must include a clear `README.md` detailing installation steps, execution commands, dependencies, and reproduction instructions.
   - Commit history must clearly reflect the team's ongoing development effort.
   - If you choose to keep your repository private, please share access with us using our GitHub handles: `shubhambagwari` and `divyaanshmertia`
2. **2-Page Descriptive Report (PDF):**
   - Strictly limited to **2 pages** (excluding references).
   - Must rigorously define the RL formulation:
     - **State Space ($\mathcal{S}$):** Representation and dimensionality of observations.
     - **Action Space ($\mathcal{A}$):** Discrete or continuous actions available to the agent.

  - **Reward Function ($\mathcal{R}$):** Mathematical formulation and incentive design.
  - **Environment & Algorithm:** Simulation dynamics, baseline comparisons, and choice of RL algorithm (e.g., Q-Learning, DQN, PPO).
- Include relevant graphs.
- Keep your GitHub link, team member/s at the top of your report. The report should be in a two-column format (eg: Overleaf LaTeX).

3. **Notes on Evaluation:**
   - Marks will be awarded relative to the performance of other teams.
   - Bonus marks will only be awarded for highly unique ideation and exceptional effort.

Strict Policy on Plagiarism & Code Copying

1. **No GitHub Forking / Copying:** Directly cloning, copying, or slightly modifying existing open-source RL repositories or public projects is strictly prohibited.
2. **Strict Action on Plagiarism:** All submitted repositories and reports will undergo thorough automated code similarity checks and manual code inspection. If any team is caught copying code from external sources, another student, or submitting an unoriginal project, **the team will receive an immediate zero (0) in AI Assignment 2 (AI-A2) and will be reported for disciplinary action.**
3. **Deadlines:** Failure to submit either the proposal or the final deliverables within the designated windows will result in an immediate zero (0).